The Karamlou–Shah no-go is not an H^2 obstruction: a level analysis, and the correct scope of the cohomological closure theorem

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Abstract

The companion note [1] classifies the global-closure obstruction (G) to a strict factorization $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ as a cohomology class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$. A natural and tempting conjecture — recorded as a prove target — asserted that the Karamlou–Shah no-go theorems (the list, stream, and tree directed containers do *not* distribute over the distribution monad Δ or the nondeterminism monad P) are *instances* of this obstruction: (G) failures witnessed by a nontrivial $[\omega]$. We *refute* this conjecture, and we do so in a way that sharpens rather than weakens the program. The refutation is a *level theorem*. The cohomological criterion is defined only when the prescribed right factor $\mathcal{D}$ is a small category — equivalently (Ahman–Uustalu) a directed container, equivalently a polynomial comonad. We prove by elementary means that the finite powerset monad P and the distribution monad Δ are *not* polynomial functors; hence they cannot be presented as a wide subcategory $\mathcal{D}$, and the crucial step "express the monad as $\mathcal{D}$" is unsatisfiable. The Karamlou–Shah obstruction therefore lives strictly *below* the local-freeness condition (L): it is a non-representability (failure-to-be-a-directed-container) phenomenon in the wrong bicategory, not a cohomological closure phenomenon. We make the bicategorical mismatch precise — their distributive law is a *mixed* comonad–monad law in $[\mathbf{Set}, \mathbf{Set}]$ (an entwining/lifting), whereas H^2 classifies distributive laws of two *categories* in $\mathbf{Span}(\mathbf{Set})$ — and we delineate exactly the salvageable positive content: among genuine *directed-container-over-directed-container* no-go theorems, the obstruction *is* a class in $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$, realized by the rigid twist as the generator of $\mathbb{Z}/2$. The correct grant statement is the scoped one. All numerical claims are machine-checked (`ks_level_check.py`, `cohomology_holonomy.py`).

1 The conjecture, and what would make it true

The prove target read, verbatim in its success criteria:

1. Express list/stream/tree directed containers as small categories $\mathcal{C}$.
2. *Express probability/nondeterminism monad as a wide subcategory $\mathcal{D}$.*
3. Verify or refute condition (L).
4. Compute the cocycle ω and show $[\omega] \neq 0$ in $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$.

Steps 3 and 4 presuppose step 2. The whole machinery of [1, 2] — hom-presheaves $\mathrm{Hom}_{\mathcal{K}}(-, b)$ as right $\mathcal{D}$-sets, free bases, transversals, the defect 2-cochain, the complex $C^\bullet(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ — is *defined relative to a wide subcategory $\mathcal{D} \subseteq \mathcal{K}$.* If the monad cannot be a wide subcategory, the conjecture does not typecheck, let alone hold. So the entire weight of the conjecture rests on step 2. We show step 2 is impossible.

The only route to step 2. A monad $T\colon \mathbf{Set} \to \mathbf{Set}$ "is" a category in exactly one way relevant here: through the equivalence

$$\text{directed container} \;\simeq\; \text{small category} \;\simeq\; \text{polynomial comonad}$$

of Ahman–Chapman–Uustalu and Ahman–Uustalu [3, 4]. A small category $\mathcal{D}$ presents the polynomial endofunctor $X \mapsto \coprod_{a\in \mathrm{ob}\,\mathcal{D}} X^{\mathcal{D}(a,-)}$ ("shapes = objects, positions = outgoing arrows"), with its canonical comonad structure. *This is the sense in which the right factor $\mathcal{D}$ of a Zappa–Szép product is an endofunctor of* **Set**. For the Karamlou–Shah question "does the directed container distribute over T" to become an instance of a strict factorization, the monad T ($= \Delta$ or P) must be realized as such a $\mathcal{D}$ — i.e. T must be a polynomial functor. (The orientation is irrelevant: whichever of $\mathcal{C}, \mathcal{D}$ one assigns to T, that factor must be a small category, hence T must be polynomial. See Remark 5.)

We now prove the route is blocked.

2 P and Δ are not polynomial functors

Recall a functor $F\colon \mathbf{Set} \to \mathbf{Set}$ is *polynomial* (a *container*) iff it is a coproduct of representables, $F(X) \cong \coprod_{s\in S} X^{P_s}$ for a set S of shapes and position-sets P_s; equivalently iff F preserves connected limits (wide pullbacks) [5, 6]. Two elementary invariants suffice here: $F(1) \cong S$ and $F(0) \cong \{s : P_s = \varnothing\}$.

Proposition 1 (Powerset is not polynomial)**.** *The finite (or full) covariant powerset functor* P *is not polynomial.*

Proof. Suppose $\mathsf{P} \cong \coprod_{s\in S}(-)^{P_s}$. Evaluating: $\mathsf{P}(1) = 2$ forces $|S| = 2$. $\mathsf{P}(0) = 1$ (the only subset of $\varnothing$ is $\varnothing$) forces *exactly one* shape to have an empty position-set; call the other shape's position-set size $k \geq 1$. Then

$$\mathsf{P}(2) \;=\; 2^0 + 2^k \;=\; 1 + 2^k.$$

But $\mathsf{P}(2) = 4$ (the four subsets of a two-element set), so $2^k = 3$, which has no solution in integers k. Contradiction. ☐

Proposition 2 (Distribution is not polynomial)**.** *The finitely-supported distribution monad* Δ *is not polynomial.*

Proof. $\Delta(1) = 1$ (the unique distribution on a point) forces $|S| = 1$ if Δ were polynomial, i.e. $\Delta \cong (-)^P$ would be *representable*. A representable functor preserves all limits, in particular finite products: $(-)^P$ sends $A \times B$ to $A^P \times B^P = (A^P) \times (B^P)$, and the comparison $(A\times B)^P \to A^P \times B^P$ is the identity (an iso). We exhibit that the corresponding comparison for Δ — the marginal map $\langle \Delta\pi_A, \Delta\pi_B\rangle\colon \Delta(A \times B) \to \Delta(A) \times \Delta(B)$ — is not injective for $A = B = 2$. Take the two joint distributions on $\{0, 1\} \times \{0, 1\}$

$$\mu_{\text{ind}} = \tfrac{1}{4}\delta_{(0,0)} + \tfrac{1}{4}\delta_{(0,1)} + \tfrac{1}{4}\delta_{(1,0)} + \tfrac{1}{4}\delta_{(1,1)}, \qquad \mu_{\text{cor}} = \tfrac{1}{2}\delta_{(0,0)} + \tfrac{1}{2}\delta_{(1,1)}.$$

Both have uniform marginals $(\frac{1}{2}, \frac{1}{2})$ on each factor, yet $\mu_{\text{ind}} \neq \mu_{\text{cor}}$. So the marginal map is non-injective, Δ does not preserve the product 2×2, hence is not representable, hence (being 1 on a singleton) not polynomial. ☐

Both computations are confirmed in `ks_level_check.py`: for P, $1 + 2^k = 4 \Rightarrow 2^k = 3$ has no integer solution; for Δ, the two displayed joints have equal marginals and are distinct.

Remark 3 (Why this is the heart of the matter)**.** P and Δ are the two canonical *non-polynomial* **Set**-monads — P fails to preserve wide pullbacks, Δ fails to preserve products. The entire directed-container / **Poly** / $\mathbf{Cat}^{\#}$ world in which our H^2 obstruction is defined is, by construction, the world of *familially representable* structure [7]. The Karamlou–Shah monads are precisely the standard inhabitants of its complement. The conjecture asked for the obstruction in a world the obstruction cannot see.

3 The level theorem

Theorem 4 (Level theorem)**.** *Let the right factor of a sought strict factorization be a small category $\mathcal{D}$, as required by the criterion of [2, 1]. Then:*

1. (Negative.) *The Karamlou–Shah right factors Δ and P cannot be presented as such a $\mathcal{D}$: by Propositions 1–2 they are not polynomial functors, hence not directed containers, hence not small categories acting by precomposition on* **Set**. *Consequently success-criterion* 2 *is unsatisfiable; conditions* (L) *and* (G) *are not even* defined *for these monads, and the proposed identification "K–S no-go = (G) failure = $[\omega] \neq 0$" does not typecheck. The Karamlou–Shah obstruction lies strictly* below (L)*: it is a* representability *obstruction in the bicategory* $[\mathbf{Set}, \mathbf{Set}]$*, prior to any cohomological closure datum.*

2. (Positive.) *When both factors* are *directed containers (small categories), the existence question for a complementary $\mathcal{C}$ with $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ is governed, under hypothesis* (H) *of [1], by $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$, and there are genuine* no-go *instances with $[\omega] \neq 0$. The rigid twist of [1, Ex. 5.1] realizes the generator of $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$.*

Proof. (1) Immediate from Propositions 1–2 together with the equivalence directed container $\simeq$ small category $\simeq$ polynomial comonad [3, 4]: a wide subcategory $\mathcal{D} \subseteq \mathcal{K}$ is in particular a small category, whose induced **Set**-endofunctor is polynomial; Δ, P are not polynomial, so no wide subcategory induces them. The conditions (L), (G) are predicates on the pair $(\mathcal{K}, \mathcal{D})$ with $\mathcal{D}$ a wide subcategory, so they have no value when $\mathcal{D}$ does not exist.

(2) Is Theorem 5.2 (the rigid twist) of [1], recalled in §5: there $\mathcal{D} = \mathcal{E}$ is the endomorphism subcategory, a directed container ($\mathbb{Z}/2$ at a, trivial elsewhere); (L) holds; (G) fails; $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$ and the defect represents its nonzero element. Both factors $\mathcal{C}, \mathcal{D}$ are small categories, hence directed containers. ☐

Remark 5 (Orientation independence)**.** One might hope to dodge Theorem 4(1) by assigning the monad T to the *left* factor $\mathcal{C}$ instead of the right factor $\mathcal{D}$. This does not help: in a strict factorization $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ *both* $\mathcal{C}$ and $\mathcal{D}$ are wide subcategories of $\mathcal{K}$, hence both are small categories, hence both induce polynomial endofunctors. Realizing $T = \Delta$ or P as either factor requires T polynomial. The obstruction of Propositions 1–2 is symmetric in the two factors.

4 The bicategorical mismatch

The deepest way to see why the conjecture mis-fires is that it silently translates between two different notions of distributive law living in two different bicategories.

(A) The Zappa–Szép law (where H^2 lives). A strict factorization $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ is the same datum as a distributive law of *categories* $\lambda\colon \mathcal{DC} \Rightarrow \mathcal{CD}$ — a matched pair [2, §4]. Regarding a small category as a monad in the bicategory **Span**(**Set**) on its object set, this is a distributive law of *two monads in* **Span**(**Set**), and the classical correspondence "distributive laws $\leftrightarrow$ strict factorizations" (Rosebrugh–Wood [8], Street [9]) applies. Both participants are monads (categories). The obstruction $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ measures the failure of a transversal of $\mathcal{C}$-generators to close.

(B) The Karamlou–Shah law (what they actually study). A directed container is a *comonad* G on **Set**. "G distributes over the monad T" is a *mixed* distributive law $\lambda\colon GT \Rightarrow TG$ (or its companion $TG \Rightarrow GT$) in the endofunctor bicategory $[\mathbf{Set}, \mathbf{Set}]$ — a comonad and a monad, not two monads. Such a law is exactly an *entwining/lifting* datum (Beck [10]; Brzeziński–Majid [11]; Power–Watanabe [12]): it lifts T to the category of G-coalgebras and lifts G to the category of T-algebras, and its (co)algebras are *bialgebras*. It does *not* present any category $\mathcal{K}$ as a factorized $\mathcal{C} \bowtie \mathcal{D}$; the composite GT is in general neither a monad nor a comonad. So (B) is not an object that the H^2 theorem of (A) classifies, *independently* of whether T is polynomial.

Thus there are two independent reasons the conjecture fails, and they reinforce each other: *(i)* $T = \Delta, \mathsf{P}$ is not even a category, so cannot be a factor (Theorem 4(1)); and *(ii)* even a perfectly good comonad–monad law is the wrong *kind* of distributive law — a lifting, not a factorization. The cohomological obstruction answers a question (existence of a complementary wide subcategory) that the Karamlou–Shah setup never poses.

Remark 6 (Two near-misses, ruled out)**.** *(Kleisli/Lawvere presentation.)* The one way to make T "a category" other than as a directed container is to take its Kleisli category $\mathrm{Kl}(T)$ (so $\mathrm{Kl}(\mathsf{P}) = \mathbf{Rel}$, $\mathrm{Kl}(\Delta) =$ stochastic maps) or its Lawvere theory. Neither yields a Zappa–Szép instance. A wide subcategory must share the object set of $\mathcal{K}$; $\mathrm{Kl}(T)$ has *all sets* as objects and is not small, while a directed container $\mathcal{C}$ has the shape set (e.g. $\mathbb{N}$ for stream/list) as objects — there is no common $\mathcal{K}$ in which both are wide. Moreover the question "does $\mathcal{C}$ admit a law over $\mathrm{Kl}(T)$" is again a lifting question (case (B)), not a factorization. So this route reproduces both obstructions rather than escaping them.

5 What survives: directed-container no-go theorems *are* H^2

The positive half of Theorem 4 is the grant-relevant residue, and it is genuinely a "no-go theorem with a computable cohomological obstruction" — only for the right class of monads.

Example 7 (A directed container that does not distribute, with obstruction $\mathbb{Z}/2$)**.** Objects a, x, y; $\mathrm{End}(a) = \{\mathrm{id}_a, g\}$ with $g^2 = \mathrm{id}_a$, other endomorphism monoids trivial; $\mathrm{Hom}(a, x) = \{p, pg\}$, $\mathrm{Hom}(a, y) = \{q, qg\}$, $\mathrm{Hom}(x, y) = \{s, s_2\}$ with the *rigid twist* $s \circ p = q,\ s_2 \circ p = qg$. Take $\mathcal{D} = \mathcal{E}$, the endomorphism subcategory: a directed container, equal to $\mathbb{Z}/2$ concentrated at a. Then (L) holds (every hom-set is free over its source vertex group), hypothesis (H) holds (abelian vertex groups, trivial left action), and

$$H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2, \qquad [\omega] \neq 0,$$

so no complementary directed container $\mathcal{C}$ makes $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$: the directed container $\mathcal{E}$ admits no Zappa–Szép complement inside $\mathcal{K}$. The four valid transversals all yield a defect outside B^2 representing the same nonzero class (machine-checked, `cohomology_holonomy.py`: $\dim_{\mathbb{F}_2} H^2 = 1$, no closing transversal).

This is the honest analogue, *inside the polynomial world*, of a Karamlou–Shah statement: a directed container (here $\mathcal{E}$) for which a desired distributive law (here the matched pair assembling $\mathcal{K}$) provably does not exist, *and* whose obstruction is an explicit, computable cohomology class. The contrast with Karamlou–Shah is exactly the content of Theorem 4: when the second factor is itself a directed container, the obstruction is the H^2 class; when it is Δ or P, there is no second factor at all and the failure is one level deeper.

6 Honest scope

We have *not* reproved the Karamlou–Shah no-go theorems (that no mixed distributive law $GT \Rightarrow TG$ exists for G the list/stream/tree container and $T = \Delta, \mathsf{P}$); that is a theorem about liftings in $[\mathbf{Set}, \mathbf{Set}]$, established by them and not reconsidered here (we did not have access to their proof, and worked only from the published statement: the monads are Δ and P). What we have established is sharper than a reproof: the *type* of their obstruction. It is *not* the cohomological closure class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$; it cannot be, because their second factor is not a directed container and their law is not a factorization. The two facts we contribute — the elementary non-polynomiality of P, Δ (Props. 1–2) and the bicategorical distinction between mixed laws and factorizations (§4) — are each independently sufficient to refute the conjectured identification, and both are machine- or hand-checkable.

Consequence for the grant narrative. The slogan "compositional correctness has a computable obstruction" stands, correctly scoped: *for directed-container distributive laws — the polynomial / $\mathbf{Cat}^{\#}$ world — the obstruction to a strict factorization is the computable class $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$, vanishing iff a complement exists.* The probability and nondeterminism no-go theorems are a *different* phenomenon, a non-representability obstruction outside this world, and conflating the two would have been an overclaim. Locating the boundary precisely is the result.

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